Programming Fundamentals II: Lab 02

Due Date: Tuesday September 17, 2024 at Midnight

Part 1. Intro to (Multi-Dimensional) Arrays (10 Marks)

For this week's lab, we are going to work with the basics of 2-dimensional arrays in Java.

We'll continue working with data pulled from Google's AdWords API. For this lab, we will be improving our program for doing basic data analysis on this data. Here's the data we'll be using this time:

keyword	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23	Jan-24	Feb-24	Mar-24	…
java	201000	200000	200000	201000	201000	246000	450000	201000	…
c++	60500	74000	74000	60500	49500	74000	74000	60500	…
sql	135000	135000	135000	135000	110000	165000	135000	135000	…
javascript	201000	201000	201000	165000	165000	201000	201000	165000	…
c#	40500	40500	33100	33100	27100	40500	33100	33100	…

Data Source: Google Adwords API, Sep 01, 2024

Download the Lab02.java file provided. In this file, we have our calcAverage and calcMax methods from the first lab. Additionally, we have 3 declared arrays: One with the names of the months, one with the names of our search terms, and a 2D array with the search volumes.

Note: This is just sample, our solutions will need to work even if we were to add more search terms, or more months to our data.

Make the following changes to this file:

a. **(1 Mark)** For each search term, display the average search volume for that search term, as well as the name of the month that that search term had the highest value for. *Do not modify the existing calcAverage and calcMax methods.*
b. **(2 Marks)** Create a method named calcMonthMax. This method should accept in a 2D array of search volumes, as well as the index of a month. This method will return *the index* of the search term that had the greatest search volume for that month. (If there's a tie for greatest, return the first value that tied.)
c. **(2 Marks)** Create a method named calcMonthTotal. This method should accept in a 2D array of search volumes, as well as the index of a month. This method will return the total search volume across all terms for the given month.
d. **(2 Marks)** Create a method called calcGrandTotal. This method will accept our 2D array of search volumes and return the grand total of all the search volumes.
e. **(2 Mark)** For each month, display the *name of the search term* that had the greatest search volume that month, and the total search volume for that month.
f. **(1 Marks)** Display the grand total search volume (the total of all months and search terms).

Add the **date and your name and student number** at the top of the java file. Please submit your Lab02.java file to Nexus before midnight on the due date. **Late submissions will be not be accepted.**